

**Part A: Multiple Choice (25\*2 Pts = 50 Pts)**

1-5 DCAED                      6-10 BBAAD                      11-15 CEBBB  
 16-20 DDBBB                      21-25 DBDAB

**Part B: Short Answer (5\*4 Pts = 20 Pts)****B1 (4 Pts)**

- (a) Br:  $[\text{Ar}]4s^23d^{10}4p^5$     1 unpaired electron  
 (b) Hf:  $[\text{Xe}]6s^24f^{14}5d^2$     2 unpaired electrons

**B2 (4 Pts)**

- (a)  $\text{O}^{2-}$  is larger. Because the increase in electron repulsion that accompanies addition of an electron causes the electron cloud to expand.  
 (b)  $\text{K}^+$  is larger. Because the two ions are isoelectronic and  $\text{Ca}^{2+}$  has the larger  $Z$  and  $Z_{\text{eff}}$ .

**B3 (4 Pts)**

- (a)  $\text{BrO}^-$                       (b)  $\text{BrO}^-$                       (c)  $\text{HPO}_4^{2-}$                       (d)  $\text{F}^-$

**B4 (4 Pts)**

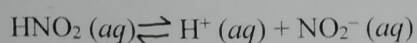
- (a) (2 Pts)  $\lambda = hc/E = (6.626 \times 10^{-34} \text{ J-s} \times 2.998 \times 10^8 \text{ m-s}) / (5.25 \times 10^{-19} \text{ J})$   
 $= \underline{3.78 \times 10^{-7} \text{ m}}$     OR    378 nm  
 (b) (2 Pts)  $\lambda = h/(mv) = (6.626 \times 10^{-34} \text{ J-s}) / (2.0 \text{ kg} \times 50 \text{ m/s}) = \underline{6.6 \times 10^{-36} \text{ m}}$

**B5 (4 Pts)**

- (a) solubility of AgI in pure water  
 $[\text{AgI}] = (K_{sp})^{1/2} = (8.3 \times 10^{-17})^{1/2} = \underline{9.1 \times 10^{-9} \text{ M}}$   
 (b)  $K = K_f \times K_{sp} = 1.0 \times 10^2 \times 8.3 \times 10^{-17} = \underline{8.3 \times 10^{-4}}$ .

**Part C: Long Answer (3\*10 Pts = 30 Pts)****C1 (10 Pts)**

- (a) (4 pts.)    Rate =  $k [\text{NO}_2]^2$   
 (b) (3 pts.)     $k = \text{Rate}/[\text{NO}_2]^2 = (1.60 \times 10^{-4}) / (0.200)^2 = \underline{4.00 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}}$ .  
 (c) (3 pts.)     $t_{1/2} = 1/(k[\text{A}]_0) = 1 / (4.00 \times 10^{-3} \times 0.400) = \underline{625 \text{ s}}$

**C2 (10 Pts)****(a) (2 Pts)****(b) (2 Pts)**

$$K_a = 4.5 \times 10^{-4}$$

At equilibrium,  $\Delta G^\circ = -RT \ln K_a = 8.314 \times 298 \times \ln(4.5 \times 10^{-4}) \div 1000 = \underline{19.1 \text{ kJ}}$

(c) (3 Pts)

At equilibrium,  $\Delta G = 0$

(d) (3 Pts)

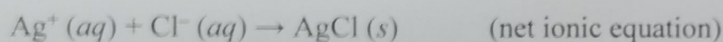
$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta G = 19.1 + 8.314 \times 298 \times \ln(5.0 \times 10^{-2} \times 6.0 \times 10^{-4} \div 0.20) \div 1000 = \underline{-2.8 \text{ kJ} < 0}$$

So the forward reaction is spontaneous at 298K.

C3 (10 Pts)

(a) (2 Pts)



(b) (2 Pts)

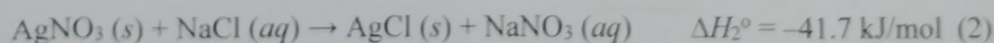
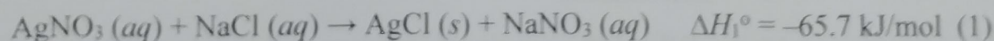
$$\begin{aligned} \Delta H^\circ &= \Delta H_f^\circ [\text{AgCl} (s)] - (\Delta H_f^\circ [\text{Ag}^+ (aq)] + \Delta H_f^\circ [\text{Cl}^- (aq)]) \\ &= -127.0 - (105.90 - 127.0) = \underline{-65.7 \text{ kJ/mol}} \end{aligned}$$

(c) (3 Pts)

Yes, I would expect  $\Delta H^\circ$  for the complete molecular equation will be the same as  $\Delta H^\circ$  for the net ionic equation.

Since the overall enthalpy change for the reaction is the enthalpy of products minus the enthalpy of reactants ( $\Delta H_{\text{rxn}}^\circ = \sum n \Delta H_f^\circ (\text{products}) - \sum m \Delta H_f^\circ (\text{reactants})$ ), the contributions of spectator ions cancel.

(d) (3 Pts)



Using Hess's law, to get  $\Delta H_3^\circ = \Delta H_2^\circ - \Delta H_1^\circ = 24 \text{ kJ/mol}$

For (3),  $\Delta H_f^\circ = \Delta H_{\text{rxn}}^\circ = \Delta H_f^\circ [\text{AgNO}_3 (aq)] - \Delta H_f^\circ [\text{AgNO}_3 (s)]$

To get,  $\Delta H_f^\circ [\text{AgNO}_3 (aq)] = \Delta H_3^\circ + \Delta H_f^\circ [\text{AgNO}_3 (s)] = 24 + (-124.4) = \underline{-100.4 \text{ kJ/mol}}$

So,  $\Delta H_f^\circ$  for  $\text{AgNO}_3(aq)$  is  $-100.4 \text{ kJ/mol}$ .